Simultaneous Scene Imaging and GMTI Scheme With Multiple Beamforming for Hypersonic Vehicle-Borne Dive-Trajectory MC-SAR

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***Abstract*— In high-resolution and wide-swath synthetic aperture radar (SAR) systems, there are azimuth Doppler ambiguity and ghost ground moving targets (GMTs). Incorporated with an improved equivalent range model (IERM), this article explores a simultaneous scene imaging and GMT indication (GMTI) scheme with multiple beamforming for hypersonic vehicle-borne dive-trajectory high-squint multichannel SAR (HSV-DTHS-MC-SAR). First, the coarse-focusing imageries are recovered. Then, the CTV estimation with the dynamic-eigenspace-based adaptive sum and difference beams (EASDBs) is explored. After that, the SAR scene processing which enjoys "customized" beamforming, minimizes the ghost GMT and reconstructs the ambiguity-free scene image. Moreover, the proposed clutter suppression method generates the beamformer center in the desired GMT direction and reduces the beamformer sidelobe by using quadratic pattern constraints. Finally, the geometry and inverse projections are performed to calibrate the distortion arising from vertical velocity. Compared to the existing methods, the proposed scheme has innovations or improvements in CTV estimation, scene recovery, and cutter suppression. To be specific, it avoids the cumbersome CTV search and thus has lower computation, enjoys more concise and robust removal of ghost GMT in the scene, and frees the steering vector mismatch plus achieves a higher signal-to-clutter-and-noise ratio (SCNR). The extensive simulations confirm the effectiveness of our proposed scheme.**

***Index Terms*— Beamforming, chirp Fourier transform (CFT), clutter suppression, dive-trajectory high-squint (DTHS), ground-moving target indication (GMTI), hypersonic vehicle (HSV), synthetic aperture radar (SAR).**

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NOMENCLATURE

AMF	Adaptive matched filtering.
ATV	Along-track velocity.
CAT	Coherent accumulation time.
CFT	Chirp Fourier transform.
CTV	Cross-track velocity.
DOF	Degrees of freedom.
DPCA	Displaced phase center antenna.
DTHS	Dive-trajectory high-squint.
EDPCA	Extended DPCA.
EASDBs	Eigenspace-based adaptive sum and difference beams.
FT	Fourier transform.
GMT	Ground moving target.
HSV	Hypersonic vehicle.
HRM	Hyperbolic range model.
HRWS	High-resolution wide-swath.
IERM	Improved equivalent range model.
STAP	Space-time adaptive processing.
ISTAP	Imaging STAP.
LCMV	Linearly constrained minimum variance.
LCMV-QPC	LCMV beamformer with quadratic pattern constraints.
LFM	Linear frequency modulation.
LRWC	Linear range walk correction.
MC-SAR	Multichannel synthetic aperture radar.
MERM	Modified equivalent range model.
PRF	Pulse repetition frequency.
RC	Range compression.
RCMC	Range cell migration correction.
SP	Subspace projection.
WAMF	Weighted AMF.

I. INTRODUCTION

THE MC-SAR [1], [2], [3], which spearheaded the evolution of HRWS imaging and GMT processing, has been published widely. However, its application on the HSV platform is still in the exploratory stage [4]. The HSV which operates in near-space (e.g., the altitude of 20–100 km), possesses the dual advantages of fast response of airborne platform and long-term observation of space-borne platform. While the HSV-MC-SAR may not replace airborne and space-borne radar systems, it offers the opportunity to bridge

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the gap between "sky" and "space" [5], [6], [7]. However, the skipping trajectory of the HSV platform leads to an extremely complex signal model, and cannot be processed using the traditional SAR/GMT indication (GMTI) methods. Fortunately, the acceleration or jerk is only present in less than one-sixth of the skipping trajectory, so a simpler signal model can be established during the uniform dive trajectory [8], [9].

In recent years, there have been several researches on HSV-SAR/GMTI. In side-looking mode, [6] and [7] suppress the stationary scene with STAP and achieve GMT imaging plus localization without ambiguity artifacts, while [8] deals with the GMTI in dive trajectory. Although the stationary scene [10] or GMT [11], [12], [13] focusing on the observation area on the squint side of the trajectory are involved, the simultaneous scene imaging and GMTI for HSV-DTHS-MC-SAR is still not available at present and thus needs further study.

Simultaneous scene imaging and GMTI for HSV-DTHS-MC-SAR has issues different from that of the classical airborne or space-borne SAR in side-looking mode [14], [15], [16], [17]. These are discussed as follows.

1) While it is safe to assume that the effect arising from the high-order phase may be disregarded during the GMT coarse- or finer-focusing for the case with the observation area on the side of the ideal orbit [18]. The existing range models of DTHS-SAR are far from adequate. Although the high-accuracy fourth-order range model [19], [20] meets the demand for high-resolution SAR imaging, its cumbersome algebraic equation limits the application of scene imaging and GMTI methods. On the other hand, the "concise" range models [21], [22], [23], [24] may not be adaptable to high-resolution ground detection. To cope with this problem, a MERM is provided in [25], but it does not take into account the GMTs. While [13] generalizes the range model to the GMTs, the processing of SAR scene imaging has not yet been realized.
2) The CTV estimation usually plays a key role in scene imaging or GMTI [26], [27]. Incorporated with specific chirp imageries, [28] introduces a feasible CTV estimation method in the mono-channel SAR case. In the case of MC-SAR, the interferometry information can be employed to implement the CTV estimation [29], [30]. Moreover, search methods such as the SP [31], the AMF [32], the WAMF [33], and the specific joint-pixel [34] methods could also improve CTV estimations. However, these methods would suffer from performance degradation due to Doppler ambiguity. Some CTV estimation methods including the CFT-search [12] and CFT-modified-search [13] methods, work well but the search processing unnecessarily increased their computational complexity. To reduce the computational time, the EASDB operation obtains accurate CTV [7] by placing zeros in the directions of clutter and its Doppler ambiguity components, performing the difference beam constraint and mitigating the pointing error sensitivity. However, its performance deterioration may not be so easily inhibited, especially for the case with high squint angle, dive trajectory, and multiple GMTs.
3) To inhibit the ghost GMT [35] in scene imagery, GMT suppression with STAP method is derived and discussed in [36], [37], and [38]. However, this method suffers from performance deterioration due to the limited DOF of HSV platform [39], [40], [41], [42], because it needs DOFs greater than twice the Doppler ambiguity number. To reduce the required DOF so as to mitigate the ghost GMT in scene imagery, the Deramp-STAP method which requires only two DOFs to suppress multiple GMTs, and yet is able to generate nulls in all-clutter directions, is proposed in [27]. However, it generates dynamic steering vectors by searching for the minimum signal-to-clutter-and-noise ratio (SCNR) of the SAR image, and the process is relatively cumbersome. Moreover, it may not be so easy to eliminate the ghost GMT and scene ambiguity components [6], [7], [8], in the case of high-speed, diving, and squint-looking HSV-SAR.
4) It is well-noted that clutter suppression is essential in the GMTI, which forms a basis for the subsequent GMT focusing and location. Generally speaking, the DPCA [43], [44], [45] method is capable of suppression the stationary clutter for the case with no channel and co-registration errors, but the ideal conditions are difficult to meet in practice, which may limit the clutter suppression performance. Then, the STAP [46], [47] method with adaptive filtering has been provided for the clutter cancellation in the MC-SAR case, but it is not suitable for weak targets due to the short CAT. Interconnections with long CAT, the EDPCA [48], [49], [50] and the ISTAP [51] methods improve energy integration, and thus perform reasonably well for the case with weak targets. However, these methods require more than twice DOF as the Doppler ambiguity number, in order to form the nulls in the directions of the all-clutter (e.g., clutter and its ambiguity components) and GMT's ambiguity components [52], [53]. To decrease the DOF requirement, the CFT [54], [55] tool has been researched and applied. To be specific, the CFT-center [18] and CFT-two-step [6] methods recover the ambiguity-free GMT, form the main beam in the GMT region, and generate zeros in the all-clutter directions. More importantly, the effects of GMT ambiguity components are mitigated by compressing the azimuthal Doppler spectrum, which means that the DOF requirement is reduced to slightly larger than the Doppler ambiguity number. However, the steering vector mismatch arising from the imprecise CTV makes it difficult to avoid GMT energy loss. Although the CFT-search [12] and CFT-modified-search [13] methods improve the matching between the main-beam peak and GMT, it greatly increase the computational complexity due to the specific search operation. While the CFT-accurate method [7] with LCMV beamformer, which avoids the search operation is able to reduce the computation time, its higher sidelobe may affect GMT extraction. To further reduce the

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sidelobe, the LCMV-QPC [56] is discussed in array radar systems, but it does not take into account the sensitivity of GMT direction [27].

To address these issues, incorporated with an IERM, a simultaneous scene imaging and GMTI scheme with multiple beamforming for HSV-DTHS-MC-SAR is derived. After the RC, LRWC, and the azimuth cubic CFT, the echoes are transformed into the RC and azimuth CFT frequency (RC-CFT) domain, and thus the coarse-focusing imageries are recovered. Then, the CTV estimation with the dynamic-EASDB is explored. After that, the SAR scene processing which enjoys "customized" beamforming, minimizes the ghost GMT and reconstructs the ambiguity-free scene image. Moreover, the GMT processing is derived, which includes the clutter rejection with matched LCMV-QPC and GMT imaging. Finally, the geometry and inverse projections are performed to calibrate the distortion arising from vertical velocity. To summarize, the proposed scheme has several attributes as follows.

1) The coarse-focusing preprocessing which uses the azimuth cubic CFT to mitigate the ambiguity GMT, is able to reduce the DOF requirement for subsequent processing in the case of sparse GMT scene, thus basically achieving the azimuth compression plus cubic RCMC.
2) The CTV estimation with dynamic-EASDB which combines high efficiency and high accuracy in multiple GMT cases, forms a basis for ameliorating or simplifying the subsequent scene imaging and GMTI. Moreover, it combines the concept of dynamic beamforming and extends the application to multiple GMTs.
3) The scene recovery and GMT suppression applies "customized" beamforming to form nulls in the desired directions, thus removing ghost GMT from the recovered scene, and facilitating GMT localization. In addition, it only needs two DOFs to suppress the multiple GMTs when the GMT is sparse.
4) The clutter suppression with matched LCMV-QPC enjoys higher matching and lower sidelobe, and has the higher SCNR. Moreover, the GMT can be focused and located in the scene image, because the Doppler aliasing and GMT motion impacts are mitigated or removed.

The remainder of this article is structured as follows. In the HSV-DTHS-MC-SAR case, IERM is provided in Section II. A simultaneous scene imaging and GMTI scheme with multiple beamforming is introduced in detail in Section III. In Section IV, extensive simulations and examples of our proposed scheme are provided. Finally, conclusions are drawn in Section V.

Notation: The differences between this article and [13] are summarized in Table I.

Notation: The definition of the terms is listed in Nomenclature.

II. Improved Equivalent Range Model

To simplify and improve subsequent processing of simultaneous scene imaging and GMTI, the IERM is provided in this section.

TABLE I
DIFFERENCES BETWEEN THIS ARTICLE AND [13]

Main Step	[13]	This paper
CTV estimation	Parameter search	Dynamic-EASDB
Ghost GMT suppression	/	"Customized" beamforming
Clutter suppression	LCMV	LCMV-QPC

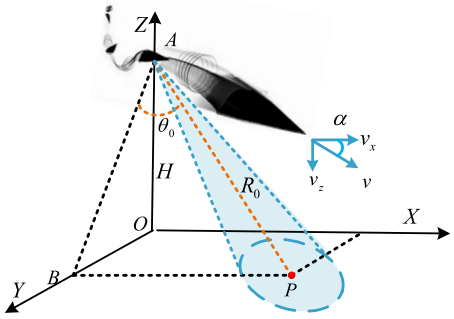


Fig. 1. Sketch of the HSV-DTHS-MC-SAR.

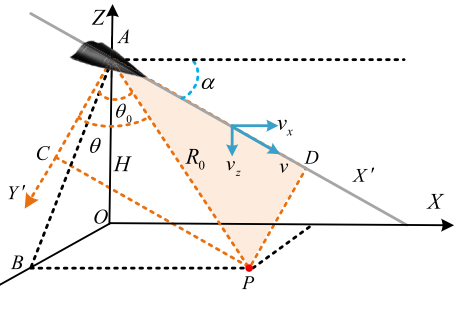


Fig. 2. Sketch of equivalent geometry.

A. Stationary Targets

The skipping trajectory of HSV can be approximated as a straight line within a synthetic aperture time, as shown in Fig. 1. The acceleration or jerk is only present in less than one-sixth of the skipping trajectory, so the range model can still be established during a uniform dive trajectory. Interconnections with the inertial navigation system, the geometry configuration of HSV-DTHS-MC-SAR is sketched in Fig. 2. Assume stationary clutter P is located at the antenna footprint center, and the parameters are defined as follows.

H Altitude of the HSV.
α Dive angle.
θ_0 Squint angle ($\theta_0 > 45°$).
R_0 Slant range of the antenna footprint center.
v Synthesized velocity of the HSV.
v_x Horizontal velocity component of the HSV along X-axis.
v_z Vertical velocity component of the HSV along Z-axis.

On the imaging plane ABP, the range model between the nth subarray and a stationary clutter can be introduced as

$$R_{\mathrm{ST}}(t_a, R_0) = \sqrt{R_0^2\cos^2\theta_0 - H^2 + (v_x t_a - R_0\sin\theta_0)^2 + (H - v_z t_a)^2} \tag{1}$$

where t_a is the azimuth time.

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Then, (1) is redefined as

$$R_{\mathrm{ST}}(t_a; R_0)=\sqrt{{R_0}^2+v^2{t_a}^2-2R_0v\sin\theta t_a} \tag{2}$$

with

$$\begin{cases} v=\sqrt{v_x^2+v_z^2} \\ v\sin\theta=v_x\sin\theta_0+Hv_z/R_0 \end{cases} \tag{3}$$

where θ represents the equivalent squint angle. The equivalent geometry of the DTHS-MC-SAR is sketched in Fig. 2, the first component of (3) represents the synthesized velocity of HSV, the term $(v_x\sin\theta_0+Hv_z/R_0)$ indicates the projection of the horizontal and vertical velocity components in the beam direction.

From (2) and (3), the IERM of a stationary clutter on new imaging plane ACPD in Fig. 2 can be shown

$$R_{\mathrm{ST}}(t_a, R_0)=\sqrt{a_0+a_1t_a+a_2{t_a}^2} \tag{4}$$

with

$$\begin{cases} a_0={R_0}^2 \\ a_1=-2R_0v\sin\theta \\ a_2=v^2. \end{cases} \tag{5}$$

Observing (4), the range model is similar to the classical HRM in squint-looking mode. The difference is that the horizontal velocity is replaced by the synthetic velocity, which means that the dive trajectory is equivalent to the horizontal trajectory in the synthesized velocity direction. By changing the imaging plane to plane ACPD with the above equivalence, the side-effect arising from vertical velocity in the range model can basically be alleviated and the translation invariance is valid in the synthesized velocity direction. In fact, the plane ACPD can be seen as a rotation of the plane ABP along the AP axis, and this rotation results in imaging distortion that does not reflect the true range plus azimuth position information and requires further geometric correction processing to recover the accurate position information, which will be discussed later.

Using three-order Taylor series, (4) can be expanded as

$$R_{\mathrm{ST}}(t_a, R_0)=b_0+b_1t_a+b_2{t_a}^2+b_3{t_a}^3+\cdots \tag{6}$$

with

$$\begin{cases} b_0=\sqrt{a_0} \\ b_1=\dfrac{a_1}{2R_0} \\ b_2=\dfrac{a_2}{2R_0}-\dfrac{a_1^2}{8R_0^3} \\ b_3=-\dfrac{a_1a_2}{4R_0^3}+\dfrac{a_1^3}{16R_0^5}. \end{cases} \tag{7}$$

B. GMTs

The geometry of GMT data collection is shown in Fig. 3, where the imaging plane is the plane ACPD in Fig. 2. Taking an arbitrary GMT $T(x_T, R_0)$ as an example, the range geometry model is formulated as follows.

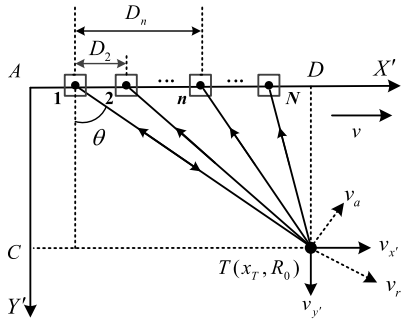


Fig. 3. Sketch of GMT data collection.

d Distance between each two neighboring channels.
M Uniform subarray containing an array element number.
N Uniform subarray numbers of a linear antenna array. $n=1,2,\ldots,N$.
D_n Distance between the transmitting channel with the nth receiving subarray, and $D_n=Md(n-1)$.
v_a ATV of a GMT.
v_r CTV of a GMT.
$v_{x'}$ X'-axis velocity component of a GMT.
$v_{y'}$ Y'-axis velocity component of a GMT.

The 2-D velocities of a GMT are

$$\begin{bmatrix} v_a \\ v_r. \end{bmatrix}=\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta. \end{pmatrix}\begin{bmatrix} v_{x'} \\ v_{y'}. \end{bmatrix}. \tag{8}$$

Form (6), the instantaneous range in terms of the nth subarray is introduced as

$$\begin{aligned} &R_{\mathrm{ST},n}(t_a, R_0) \\ &=b_0+b_1(t_a+\Delta t_n)+b_2(t_a+\Delta t_n)^2+b_3(t_a+\Delta t_n)^3 \end{aligned} \tag{9}$$

where $\Delta t_n=D_n/v$.

Considering the 2-D velocity of a GMT, the IERM on the imaging plane ACPD is defined as

$$\begin{aligned} &R_{T,n}(t_a, R_0) \\ &=c_0+c_1\left(t_a+\Delta t'_n\right)+c_2\left(t_a+\Delta t'_n\right)^2+c_3\left(t_a+\Delta t'_n\right)^3 \end{aligned} \tag{10}$$

with

$$\begin{cases} c_0=b_0 \\ c_1=-v_{\Delta r} \\ c_2=\dfrac{v_{\Delta a}^2}{2R_0} \\ c_3=\dfrac{v_{\Delta r}v_{\Delta a}^2}{2R_0^2} \end{cases} \tag{11}$$

where $\Delta t'_n=D_n/(v-v_x)$, $v_{\Delta a}=-v\cos\theta+v_a$ denotes the velocity deviation between the HSV and a GMT in along-track direction, and $v_{\Delta r}=v\sin\theta+v_r$ denotes the velocity deviation between the HSV and a GMT in cross-track direction. After the above derivation, the dive trajectory is simplified to the equivalence of a horizontal trajectory in the synthesized velocity direction, and thus the IERM with a GMT of (10) avoids the relatively cumbersome range model in [8].

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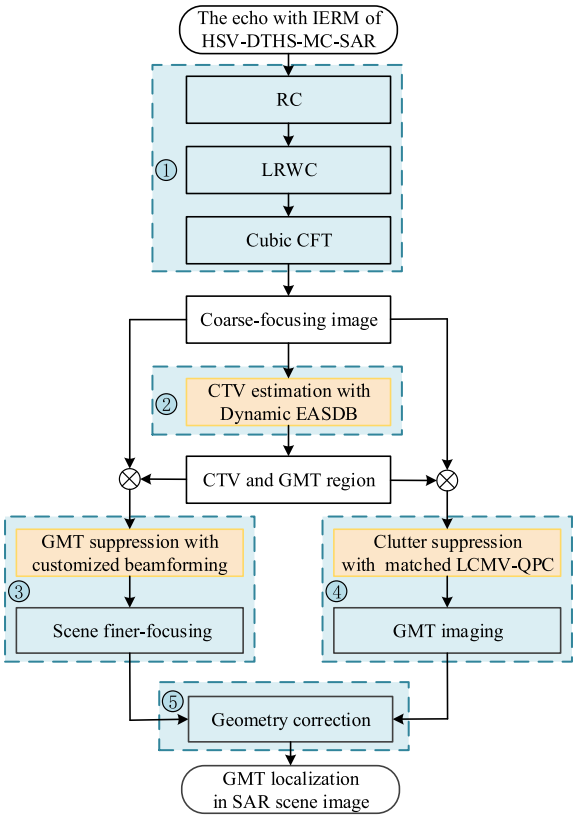


Fig. 4. Overview of the proposed scene imaging and GMTI scheme: 1) coarse-focusing preprocessing containing RC, LRWC, and cubic CFT; 2) CTV estimation with dynamic EASDB; 3) SAR scene processing with GMT suppression and scene finer-focusing; 4) GMT processing including clutter suppression and GMT imaging; and 5) geometry correction.

III. Simultaneous Scene Imaging and GMTI Scheme With Multiple Beamforming for HSV-DTHS-MC-SAR

Here, a scene imaging and GMTI scheme with multiple beamforming for HSV-DTHS-MC-SAR is derived. The overview is sketched in Fig. 4, and the whole scheme can be grouped into five blocks. First, a coarse-focusing preprocessing is performed to relieve the influences of ambiguity GMTs and cubic phase. Second, a CTV estimation with dynamic EASDB is presented, which forms a basis for improving and simplifying the subsequent scene imaging and GMTI. Third, the SAR scene processing, which enjoys GMT suppression with "customized" beamforming, minimizes the ghost GMT and reconstructs the ambiguity-free scene image. Fourth, the GMT processing which constructs a matched LCMV-QPC beamforming to suppress ambiguity clutter, achieves accurate GMT extraction. Finally, the geometry correction which calibrates the side-effect arising from vertical velocity, frees the image distortion. Note that the three yellow block diagrams in the flowchart correspond to the multiple beamforming of the proposed scheme.

A. Coarse-Focusing Preprocessing

To achieve the coarse-focusing and relieve the influences of Doppler ambiguity, the coarse-focusing preprocessing, containing RC, LRWC, and cubic CFT, is introduced. It basically achieves the azimuth compression plus cubic RCMC, mitigates the effect of ambiguity GMT, and reduces the required channel number for the subsequent CTV estimation, scene processing, and GMT processing.

Assuming that the antenna transmits LFM pulses, the GMT signal in the time domain received at the nth subarray is given by

$$\begin{aligned} S_{T0,n}(\hat{t}, t_a) = \sigma_T w_r\left(\hat{t} - \frac{2R_{T,n}(t_a, R_0)}{c}\right) w_a(t_a + \Delta t_n) \\ \times \exp\left\{ j\pi\gamma\left(\hat{t} - \frac{2R_{T,n}(t_a, R_0)}{c}\right)^2 \right\} \\ \times \exp\left\{ -j\frac{4\pi}{\lambda} R_{T,n}(t_a, R_0) \right\} \end{aligned} \tag{12}$$

where σ_T is the scattering point altitude of the GMT, $\hat{t}$ is the range-time, and γ is the chirp rate in range direction. $w_r(\cdot)$ is the range profile in time-domain, $w_a(\cdot)$ is the azimuth profile in time-domain, λ is the radar wavelength in meters, and c is the velocity of propagation.

To construct the GMT steering vector for the Doppler aliasing of echoes, GMT coarse-focusing expression in terms of the Nyquist band is rewritten as

$$\begin{aligned} S_{T,n}(\hat{t}, f_b) = \sum_{l=-L}^{L} \sigma_T G_r G_{az} w_r\left(\hat{t} - \frac{2R_0}{c}\right) \\ \times W_a\left(f_b + f_{\mathrm{dc_}b} + (l+K)\cdot f_{\mathrm{prf}}\right) \\ \times \exp\left\{ -j\frac{4\pi}{\lambda} R_0 \right\} \\ \times \exp\left\{ -j2\pi\left(f_b + l\cdot f_{\mathrm{prf}} + 2f_{\mathrm{dc}}\right)\Delta t_n \right\} \end{aligned} \tag{13}$$

where the baseband frequency is defined as $f_b = f_a - l\cdot f_{\mathrm{prf}} \in [-f_{\mathrm{prf}}/2, f_{\mathrm{prf}}/2]$, f_a is the CFT frequency in azimuth, $f_{\mathrm{dc}} = 2v_r/\lambda$ is the Doppler deviation induced by CTV, and the baseband frequency is represented by $f_{\mathrm{dc_}b} = f_{\mathrm{dc}} - K_{\mathrm{dc}}\cdot f_{\mathrm{prf}} \in [-f_{\mathrm{prf}}/2, f_{\mathrm{prf}}/2]$. One supposes that the Doppler ambiguity number $K = B_a/f_{\mathrm{prf}}$ is an odd number, e.g., $K = 2L+1$. In the case of K is an even number and $K = 2L$, the upper limitation of the integral in (13) should be changed from L to $L-1$.

One can observe that only the last component on the right side of (13) includes the subarray information. By using the following steering vector, in theory, the GMTs and stationary clutter can hence be distinguished, i.e.,

$$\mathbf{a}_T^l(v_r) = \begin{bmatrix} \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}} + 4v_r/\lambda\right)\frac{D_1}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}} + 4v_r/\lambda\right)\frac{D_n}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}} + 4v_r/\lambda\right)\frac{D_N}{v}\right) \end{bmatrix}. \tag{14}$$

By placing the velocity of GMT to zero, we construct the steering vector of stationary clutter, yielding

$$\mathbf{a}_C^l = \begin{bmatrix} \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}}\right)\frac{D_1}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}}\right)\frac{D_n}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\mathrm{prf}}\right)\frac{D_N}{v}\right) \end{bmatrix}. \tag{15}$$

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It is safe to assume that the GMT is sparse with respect to the stationary scene, and thus the Doppler aliasing of GMT is directly mitigated by applying the cubic CFT [6], [7], [12], [18]. Moreover, the clutter aliasing is present in each channel, it is necessary to place the zero in the clutter and its ambiguity component directions during CTV estimation, this will be introduced in detail in Section III-B.

B. CTV Estimation With Dynamic-EASDB

After the aforementioned coarse-focusing preprocessing, the effects of cubic RCM and GMT ambiguity are mitigated. In this section, a CTV estimation algorithm with dynamic-EASDB is derived in the RC-CFT domain, which overcomes the pointing error plus Doppler ambiguity clutter, and performs the difference beam constraint. Note that we refer to the combined dynamic beamforming concept [27] and EASDB as the dynamic-EASDB.

Taking Doppler ambiguity number $K=5$ as an example, the Doppler spectrum diagram (DSD) and beamforming of the ISTAP, CFT-search, and dynamic-EASDB methods are summarized in Fig. 5, where the black and red lines represent the frequency spectrum of a stationary scene clutter and a GMT. Also, the DOF requirements of different methods are summarized in Table II. One notices that suppressing a stationary clutter requires K DOFs since the Doppler ambiguity number for these methods is K. In Fig. 5(a), the ISTAP method requires K DOFs to suppress the GMT ambiguity components because the Doppler ambiguity number of a GMT is still K. Significantly, the DOF required to suppress the GMT ambiguity components will increase significantly, especially in the case of multiple GMTs. For the CFT-search method in Fig. 5(b), the GMT coarse-focused images are obtained in the RC-CFT domain and the GMT profile is compressed into several azimuth cells, which means that only one DOF is needed to form the main beam in the desired GMT direction for estimating CTV. However, the CTV estimations of STAP and CFT-search methods are achieved by searching for the maximum SCNR, which are very time-consuming. For the dynamic-EASDB in Fig. 5(c), the sum beam center and a difference beam zero are formed in the direction of GMT region center, and the CTV can be estimated from the monopulse ratio [7]. More importantly, the CTV-estimation with the dynamic-EASDB method not only reduces the DOF requirement for rejecting GMT ambiguity components but also avoids the time and resource consumption caused by cumbersome searches.

In the case of multiple GMTs, we adopt a dynamic beamforming concept with the DSD of GMTs. Taking two GMTs as an example, the process of dynamic-EASDB is shown in Fig. 6, where the general DSD of a clutter, first GMT, and second GMT are represented by black, red, and blue dashed lines, respectively.

For the first GMT with CFT frequency f_{a1}, the black and red curves on the right side in Fig. 6 indicate the generated sum beam and difference beam. To estimate the CTV, we use two DOFs to generate the sum beam center and a difference beam zero both in the $\sin\theta_{\text{center1}}$ direction. To estimate the CTV of the second GMT with CFT frequency f_{a2}, the direction

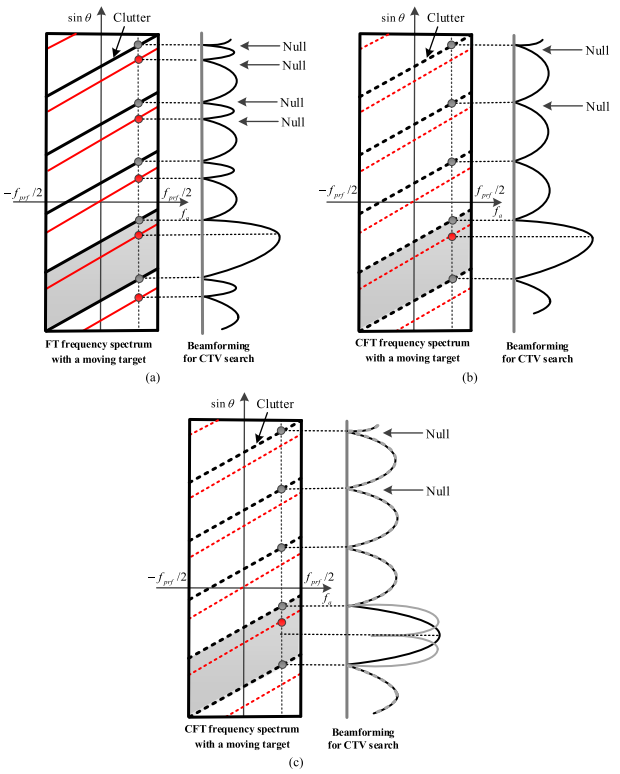


Fig. 5. CTV estimation for a GMT by applying (a) ISTAP, (b) CFT-search, and (c) dynamic-EASDB methods.

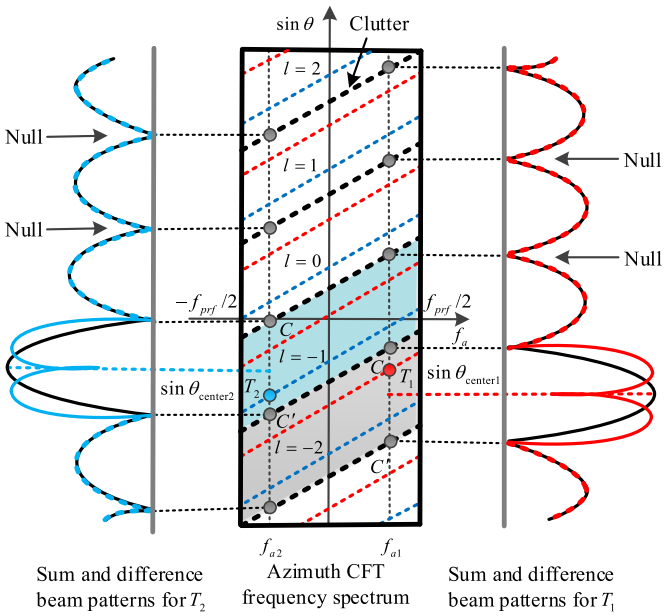


Fig. 6. CTV estimation with dynamic-EASDB for two GMTs.

of the sum and difference beam is changed from $\sin\theta_{\text{center1}}$ to $\sin\theta_{\text{center2}}$, and the generated sum and difference beams are represented the black and blue curves on the left side in Fig. 6. Incorporated with a dynamic beamforming concept, the $K+2$ DOFs are enough to recover CTV for multiple GMTs. The reason is that the coarse-focusing GMT is sparse in the observation scene and can be considered to be in different azimuth units.

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TABLE II
DOF REQUIREMENTS OF DIFFERENT METHODS IN FIG. 5

Method	DOF requirements of primary steps	Total
ISTAP [51]	Clutter suppression: K; GMT recovery: K; Main beam constraint: 1	$2K+1$
CFT-search [12]	Clutter suppression: K; Main beam constraint: 1	$K+1$
Dynamic-EASDB [7]	Clutter suppression: K; Sum and different beam constraint: 2	$K+2$

Next, taking a GMT as an example, the derivation of CTV estimation with dynamic-EASDB is discussed in detail. In Fig. 6, the first GMT, clutter, and a clutter ambiguity component are represented by T_1-point, C-point, and C'-point. The gray parallelogram region represents the first GMT region, and the midpoint of the dashed line CC' represents the GMT region center. Note that the directions of clutter, GMT, and GMT region center are reflected by $\sin\theta_C$, $\sin\theta_T$, and $\sin\theta_{\text{center}}$, and the squint angles of C-point, T_1-point, and midpoint of CC' are defined by θ_C, θ_T, and θ_{center}.

The Doppler frequency is expressed by $f_a = 2v\sin\theta/\lambda$, and the Doppler frequency of clutter, GMT, and GMT region center can be defined as

$$\begin{cases} f_a^C = 2v\sin\theta_C/\lambda \\ f_a^T = 2v\sin\theta_T/\lambda \\ f_a^{\text{center}} = 2v\sin\theta_{\text{center}}/\lambda. \end{cases} \tag{16}$$

Incorporated with general DSD, the rewritten Doppler frequency is as follows:

$$\begin{cases} f_a^C = f_b + l\cdot f_{\text{prf}} \\ f_a^T = f_b + 2f_{\text{dc}}(v_r) + l\cdot f_{\text{prf}} \\ f_a^{\text{center}} = f_b + 2f_{\text{dc}}\left(v_r^{\text{center}}\right) + l\cdot f_{\text{prf}} \end{cases} \tag{17}$$

where the CTV of GMT region center denoted by v_r^{center}, and $v_r^{\text{center}} = (f_{\text{prf}}\cdot\lambda)/4$.

The general DSD of clutter, GMT, and GMT region center can be described by

$$\sin\theta_C = \begin{cases} \dfrac{\lambda\left(f_b + L\cdot f_{\text{prf}}/2\right)}{2v} \\ \vdots \\ \dfrac{\lambda f_b}{2v} \\ \vdots \\ \dfrac{\lambda\left(f_b - L\cdot f_{\text{prf}}/2\right)}{2v} \end{cases} \tag{18}$$

$$\sin\theta_T = \begin{cases} \dfrac{\lambda\left(f_b + 2f_{\text{dc}}(v_r) + L\cdot f_{\text{prf}}/2\right)}{2v} \\ \vdots \\ \dfrac{\lambda(f_b + 2f_{\text{dc}}(v_r))}{2v} \\ \vdots \\ \dfrac{\lambda\left(f_b + 2f_{\text{dc}}(v_r) - L\cdot f_{\text{prf}}/2\right)}{2v} \end{cases} \tag{19}$$

$$\sin\theta_{\text{center}} = \begin{cases} \dfrac{\lambda\left(f_b + 2f_{\text{dc}}\left(v_r^{\text{center}}\right) + L\cdot f_{\text{prf}}/2\right)}{2v} \\ \vdots \\ \dfrac{\lambda\left(f_b + 2f_{\text{dc}}\left(v_r^{\text{center}}\right)\right)}{2v} \\ \vdots \\ \dfrac{\lambda\left(f_b + 2f_{\text{dc}}\left(v_r^{\text{center}}\right) - L\cdot f_{\text{prf}}/2\right)}{2v} \end{cases} \tag{20}$$

where the Doppler centroid of GMT region center represented by $f_{\text{dc}}(v_r^{\text{center}}) = 2v_r^{\text{center}}/\lambda$.

Inspecting (19) and (20), the calculation expression of CTV is defined as

$$v_r = v_r^{\text{center}} + (\sin\theta_T - \sin\theta_{\text{center}})v/2. \tag{21}$$

In the direction of the GMT region center, the sum beam forms a peak and the difference beam forms a null as shown in Fig. 6. From (14), the steering vector for constraining the sum and difference beams can be defined as

$$\mathbf{a}_{\text{center}}^{\hat{l}}\left(v_r^{\text{center}}\right) = \begin{bmatrix} \exp\left(-j2\pi\left(f_b + l\cdot f_{\text{prf}} + \frac{4v_r^{\text{center}}}{\lambda}\right)\frac{D_1}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\text{prf}} + \frac{4v_r^{\text{center}}}{\lambda}\right)\frac{D_n}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + l\cdot f_{\text{prf}} + \frac{4v_r^{\text{center}}}{\lambda}\right)\frac{D_N}{v}\right) \end{bmatrix}. \tag{22}$$

By selecting the maximum target magnitude from different GMT regions, a GMT region is determined. Moreover, the l is the dynamic value for CTV estimation of different GMTs (i.e., the first GMT T_1 corresponding to $l=-2$ and the second GMT T_2 corresponding to $l=-1$), as shown in Fig. 6.

With the LCMV, the beamforming is expressed as

$$\begin{cases} \min\limits_{\Omega_1(f_b)} & \Omega_1^H(f_b)\mathbf{R}(f_b)\Omega_1(f_b) \\ s.t. & \Omega_1^H(f_b) = \Phi_1 \end{cases} \tag{23}$$

where $\Omega_1(f_b)$ represents weight operator. The covariance matrix is formulated as $\mathbf{R}(f_b) = E\{\mathbf{S}(\hat{t}, f_b)\mathbf{S}^H(\hat{t}, f_b)\}$, $\mathbf{S}(\hat{t}, f_b) = [S_1(\hat{t}, f_b), \ldots, S_N(\hat{t}, f_b)]^T$ reflects the received signal after coarse-imaging processing, and the vector $\Phi_1 = [1, 0, \ldots, 0, \ldots, 0]^H$ contains $2(L+1)$ integers. $[\cdot]^T$, $E\{\cdot\}$, and $[\cdot]^H$ indicate the array operations of transpose, mean value, and conjugate transpose, respectively.

The steering vector with GMT region center direction is defined

$$\mathbf{A}\left(v_r^{\text{center}}\right) = \left[\mathbf{a}_{\text{center}}^{l}\left(v_r^{\text{center}}\right), \mathbf{a}_C^{-L}(f_b), \ldots, \mathbf{a}_C^{l}(f_b), \ldots, \mathbf{a}_C^{L}(f_b)\right]. \tag{24}$$

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From (23), the weight operator is derived, yielding

$$\Omega_1(f_b) = \mathbf{R}(f_b)^{-1}\mathbf{A}\left(v_r^{\text{center}}\right)\left(\left(\mathbf{A}\left(v_r^{\text{center}}\right)\right)^H\mathbf{R}(f_b)^{-1}\mathbf{A}\left(v_r^{\text{center}}\right)\right)^{-1}\Phi_1 \tag{25}$$

where $[\cdot]^{-1}$ represents the inverting operator.

In order to overcome the sensitivity of pointing errors and maintain the stability of the monopulse ratio curve, the echoes are transformed into the signal subspace. EASDB cancels the noise subspace components and retains the signal subspace components. Meanwhile, the constraint function mitigates the irregular beam to some extent, especially when the difference beam generates the nulls in the directions of the GMT region center and clutter. Assume that the detection region includes a GMT and P clutter. After transforming the received signal into subspace and rejecting the noise subspace, the weight operator of the sum beam is given by [7]

$$\Omega_\Sigma(f_b) = \mathbf{V}_s\mathbf{V}_s^H\Omega_1(f_b) \tag{26}$$

where the column vector of $\mathbf{V}_s = [\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_{P+1}]$ represents a signal subspace, and $\mathbf{V}_i$ indicates a feature vector.

The beamforming with difference beam constraint is as follows:

$$\begin{cases} \min\limits_{\Omega_\Delta(f_b)} & \Omega_\Delta^H(f_b)\mathbf{R}(f_b)\Omega_\Delta(f_b) \\ s.t. & \text{Re}\left(\Omega_\Delta^H(f_b)\mathbf{A}_\Delta\right) = \Theta \end{cases} \tag{27}$$

where $\Omega_\Delta(f_b)$ reflects the weight operator of difference beam and Re[·] is called the real number operator. Inspecting the constraint function of (27), the difference beam is constrained and the effects of the irregular beam are relieved, especially for the case with the nulls formed in the direction of clutter and its Doppler ambiguity components.

The steering vector of the difference beam can be described by

$$\mathbf{A}_\Delta = \left[\frac{\mathbf{A}\left(v_r^{\text{center}}+\Delta v_r\right)}{\Omega_\Sigma^H(f_b)\mathbf{A}\left(v_r^{\text{center}}+\Delta v_r\right)} \quad \frac{\mathbf{A}\left(v_r^{\text{center}}\right)}{\Omega_\Sigma^H(f_b)\mathbf{A}\left(v_r^{\text{center}}\right)} \quad \frac{\mathbf{A}\left(v_r^{\text{center}}-\Delta v_r\right)}{\Omega_\Sigma^H(f_b)\mathbf{A}\left(v_r^{\text{center}}-\Delta v_r\right)}\right] \tag{28}$$

where Δv_r denotes a velocity constant and $\Theta = [\Lambda\Delta v_r \ 0 \ -\Lambda\Delta v_r]$. Λ is the slope fit by the EASDB.

In combination with the Lagrange Multiplier algorithm, the weight operator of difference beam is derived, we have

$$\Omega_\Delta(f_b) = \mathbf{R}(f_b)^{-1}\mathbf{A}_\Delta\left[\text{Re}\left(\mathbf{A}_\Delta^H\mathbf{R}(f_b)^{-1}\mathbf{A}_\Delta\right)\right]^{-T}\Theta^T. \tag{29}$$

The direction of GMT is modeled as

$$\sin\theta_T = \text{Re}\left(\frac{S_\Delta(\hat{t}, f_b)}{S_\Sigma(\hat{t}, f_b)}\right)/\Lambda + \sin\theta_{\text{center}} \tag{30}$$

where $S_\Sigma(\hat{t}, f_b) = \Omega_\Sigma^H(f_b)\mathbf{S}(\hat{t}, f_b)$ and $S_\Delta(\hat{t}, f_b) = \Omega_\Delta^H(f_b)\mathbf{S}(\hat{t}, f_b)$ reflect the signal output obtained by sum and difference beams.

Observing (21) and (30), the CTV is introduced as

$$\hat{v}_r = v_{r0} + v\cdot\text{Re}\left(\frac{S_\Delta(\hat{t}, f_b)}{S_\Sigma(\hat{t}, f_b)}\right)/(2\Lambda). \tag{31}$$

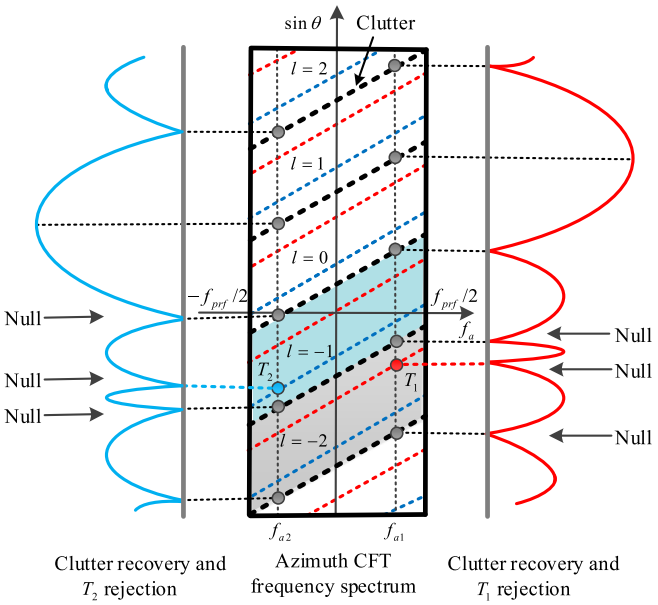


Fig. 7. Stationary scene recovery and GMT suppression with "customized" beamforming.

C. Scene Recovery and GMT Suppression With "Customized" Beamforming

With the aforementioned processing, the coarse-focusing imageries and CTV can be obtained. To recover the SAR image free from ghost GMT, the scene recovery and GMT suppression method with "customized" beamforming is proposed, which is capable of placing the zero in the expected GMT direction. Compared with Deramp-STAP method [27], the steering vector of the proposed method is directly generated from the estimated CTV and GMT region, thus avoiding the redundant search process.

The processes of stationary scene recovery and GMT suppression are sketched in Fig. 7, where the red and blue beamforming correspond to the first and second GMTs. By forming the nulls in the direction of clutter ambiguity components and GMT, one guarantees that the expected scene can be effectively recovered.

With the LCMV, the processors of ambiguity-free scene recovery and GMT suppression can be represented as

$$\begin{cases} \min\limits_{\Omega_2(f_b,\hat{v}_r)} & \Omega_2^H\left(f_b, \hat{v}_r\right)\mathbf{R}(f_b)\Omega_2\left(f_b, \hat{v}_r\right) \\ s.t. & \Omega_2^H\left(f_b, \hat{v}_r\right)\mathbf{A}\left(f_b, \hat{v}_r\right) = \Phi_2 \end{cases} \tag{32}$$

where $\Omega_2(f_b, \hat{v}_r)$ represents the weight operator for achieving the scene recovery and GMT suppression, and the vector $\Phi_2 = [0, 0, \ldots, 1, \ldots, 0]^H$ contains $2(L+1)$ integers.

The steering vector in the GMT direction of (32) is introduced as

$$\mathbf{A}\left(f_b, \hat{v}_r\right) = \left[\mathbf{a}_T^{\hat{l}}\left(\hat{v}_r\right), \mathbf{a}_C^{-L}(f_b), \ldots, \mathbf{a}_C^{l}(f_b), \ldots, \mathbf{a}_C^{L}(f_b)\right]. \tag{33}$$

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From (33), an revised steering vector in GMT region ($l = \hat{l}$) is formulated as

$$\mathbf{a}_T^{\hat{l}}(\hat{v}_r) = \begin{bmatrix} \exp\left(-j2\pi\left(f_b + \frac{4\hat{v}_r}{\lambda} + \hat{l}\cdot f_{\text{prf}}\right)\frac{D_1}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + \frac{4\hat{v}_r}{\lambda} + \hat{l}\cdot f_{\text{prf}}\right)\frac{D_n}{v}\right) \\ \vdots \\ \exp\left(-j2\pi\left(f_b + \frac{4\hat{v}_r}{\lambda} + \hat{l}\cdot f_{\text{prf}}\right)\frac{D_N}{v}\right) \end{bmatrix}. \quad (34)$$

In combination with Lagrange Multiplier algorithm, the weight operator is derived, i.e.,

$$\Omega_2(f_b, \hat{v}_r) = \mathbf{R}(f_b)^{-1}\mathbf{A}(f_b, \hat{v}_r)\left(\left(\mathbf{A}(f_b, \hat{v}_r)\right)^H\mathbf{R}(f_b)^{-1}\mathbf{A}(f_b, \hat{v}_r)\right)^{-1}\Phi_2. \quad (35)$$

Inspecting (34), by using accurate CTV to amend a GMT steering vector, it is guaranteed that GMT accumulation can be performed satisfactorily. The scene recovery processors can be defined as

$$S_{C0}(\hat{l}, f_b) = \left[\Omega_2(f_b, \hat{v}_r)\right]^H\mathbf{S}(\hat{l}, f_b). \quad (36)$$

After rejecting the scene clutter ambiguity components and GMT, the reconstructed coarse-focusing imageries without ghost GMT are obtained. The range migration algorithm (RMA) [57] or autofocus [58] methods can then be adopted to implement the finer-focusing of the SAR scene imageries.

D. Clutter Suppression With Matched LCMV-QPC

To enhance GMT amplitude and reduce the clutter amplitude so as to improve SCNR, a clutter suppression method with matched LCMV-QPC is derived. It improves the matching between the main-beam peak and GMT and thus enhances the GMT accumulation. Meanwhile, the method enhances the clutter suppression performance, because it has the ability to generate lower sidelobe and place the zeros for rejecting the ambiguity components of stationary clutter.

Fig. 8 summarizes the different clutter suppression schemes in a GMT case, where the beamformers of different methods are represented by the different color curves. These methods have the ability to extract sparse GMTs because they place the zeros in the direction of clutter ambiguity components. The proposed and CFT-accurate methods outperform the other two methods in clutter suppression and GMT extraction, because the estimated CTV is executed to generate the matched beamforming in the GMT extraction stage, and the loss of SCNR caused by beamforming mismatch can be eliminated. More importantly, one notices that the proposed method with QPC has the lower beamforming sidelobe, and thus reduces the clutter amplitude and improves the SCNR.

In the case of multiple GMTs, we adopt a dynamic beamforming concept with the DSD of GMTs. Taking two GMTs as an example, the processes of clutter rejection and GMT extraction are shown in Fig. 9. For the first GMT with CFT frequency f_{a1}, the red curve on the right side in Fig. 9 represents the matched beamforming. To extract the GMT,

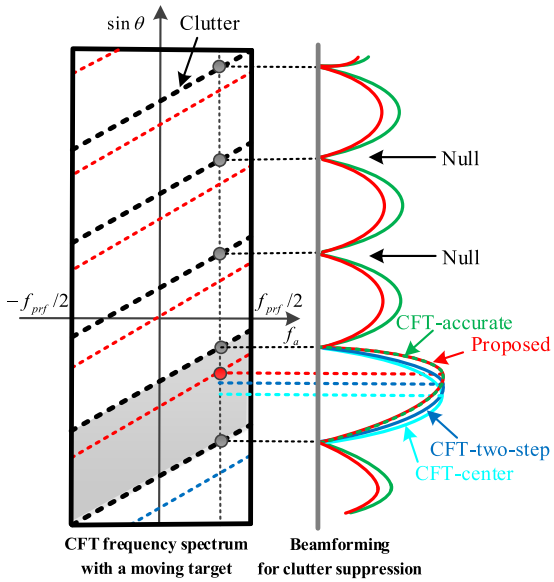


Fig. 8. Clutter suppression by applying CFT-center, CFT-two-step, CFT-accurate, and the proposed methods in a GMT case.

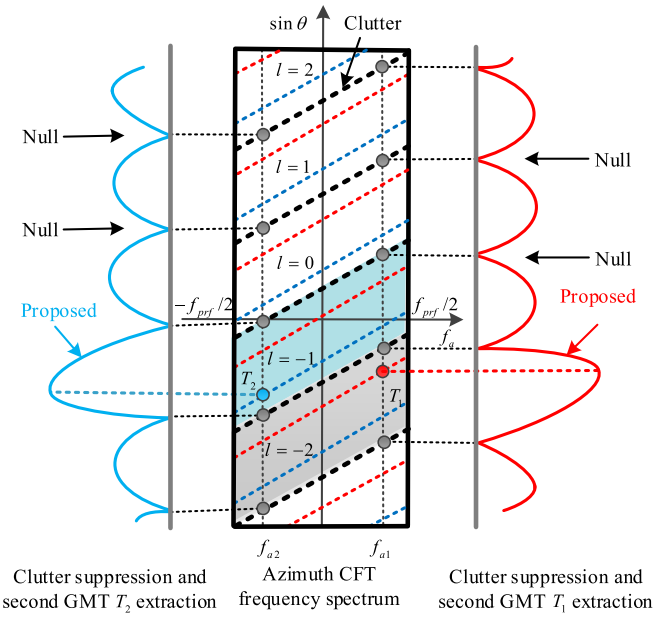


Fig. 9. Clutter suppression by using the proposed method for the case with two GMTs.

we use two DOF to generate the peak value of the main beam in the $\sin\theta_{\text{center1}}$ direction. To extract the second GMT with CFT frequency f_{a2}, the peak value of main-beam is changed from $\sin\theta_{\text{center1}}$ to $\sin\theta_{\text{center2}}$, and the matched beamforming is represented the blue curve on the left side in Fig. 9.

To improve the matching of LCMV-QPC [27], the beamforming with the GMT direction is as follows:

$$\begin{cases} \min\limits_{\Omega_3^H(f_b,\hat{v}_r)} & \Omega_3^H(f_b, \hat{v}_r)\mathbf{R}(f_b)\Omega_3(f_b, \hat{v}_r) \\ s.t. & \Omega_3^H(f_b, \hat{v}_r)\mathrm{A}(f_b, \sin\phi_{T'}) = \Phi_3 \\ s.t. & \left(\Omega_3(f_b, \hat{v}_r) - \Omega_{3,bj}(f_b)\right)^H\mathbf{Q}_j\left(\Omega_3(f_b, \hat{v}_r) - \Omega_{3,bj}(f_b)\right) \le \xi_j \end{cases} \quad (37)$$

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where Φ_3 is weight operator and $\Phi_3{=}\Phi_1$, $\Omega_3(f_b, \hat{v}_r)$ represents weight operator for achieving the clutter suppression and GMT extraction. To constrain the quadratic pattern so as to reduce the beamforming sidelobe, the original baseband Doppler frequency is divided into r areas, e.g., $f_{b1}, \ldots, f_{bj}, \ldots, f_{br}$ $(j = 1, \ldots, j, \ldots, r)$, ξ_j is the threshold of QPC. $\Omega_{3,bj}(f_b)$ represents weight operator of the beam in the $\mathbf{Q}_j$ area, $\mathbf{Q}_j = \int_{f_{bj}} \mathbf{v}(f_b)\mathbf{v}^H(f_b)\, df_b$, $\mathbf{v}(f_b)$ is the response operator of baseband Doppler frequency.

From (37), the expression of the weight operator in the GMT direction can be described as

$$\begin{aligned}&\Omega_3\left(f_b, \hat{v}_r\right)\\&= \mathbf{R}(f_b)^{-1}\mathbf{A}\left(f_b, \hat{v}_r\right)\left(\mathbf{A}^H\left(f_b, \hat{v}_r\right)\mathbf{R}(f_b)^{-1}\mathbf{A}\left(f_b, \hat{v}_r\right)\right)^{-1}\Phi_3.\end{aligned} \tag{38}$$

The clutter cancellation process can be defined as

$$S_{T0}\left(\hat{t}, f_b\right) = \left[\Omega_3\left(f_b, \hat{v}_r\right)\right]^H \mathbf{S}\left(\hat{t}, f_b\right). \tag{39}$$

Next, the GMT finer-focusing process is performed, and the side-effect of GMT's velocity and high-order phase are removed, including phase decoupling with second order keystone transform (SOKT) [6], [7], accurate RWC, ATV estimation with simplified fractional FT (SFrFT) [59], and matched filtering. The GMT finer-focusing image can be obtained, i.e., [13], [27]

$$\begin{aligned}&S_T\left(\hat{t}, t_a\right)\\&= \sigma_T' \mathrm{sinc}\left(B\left(\hat{t} - \frac{2R_0}{c}\right)\right)\mathrm{sinc}(t_a - t_c)\exp\left\{-j\frac{4\pi f_c}{c}R_0\right\}\end{aligned} \tag{40}$$

where σ_T' is the amplitude of finer-focusing GMT.

E. Geometry Correction

With the aforementioned processing, simultaneous scene imaging and GMTI of HSV-DTHS-MC-SAR are achieved. The IERM has a more concise expression and simplifies the scene imaging and GMTI procedures, due to that the dive trajectory is equivalent to the horizontal trajectory in synthesized velocity direction. However, as the imaging plane is changed into the plane ACPD, geometric distortion occurs as shown in Fig. 10. Significantly, it results in imaging results that do not reflect the true range and azimuth position information and require further geometric correction processing to recover the accurate position information. Fortunately, the geometry and inverse projections are able to calibrate the distortion. The projection expression of a target (stationary target or GMT) on the imaging plane $T(x, y)$ and ground plane $T'(x', y')$ is

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix}\begin{bmatrix} x \\ y \end{bmatrix} \tag{41}$$

$$\begin{cases} x = \dfrac{f_d \cdot \lambda R_0}{2v\cos\alpha} - H \cdot \tan\alpha \\ y = \sqrt{R_0^2 - H^2 - x^2} \end{cases} \tag{42}$$

where f_d is the target Doppler frequency on the ground plane and the azimuth angle is represented by β.

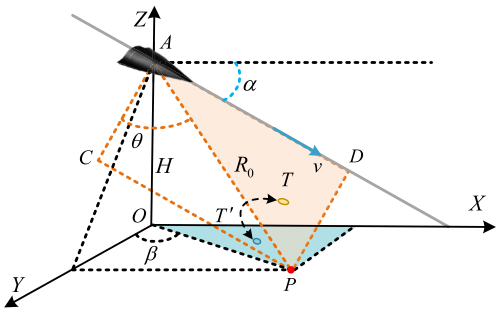
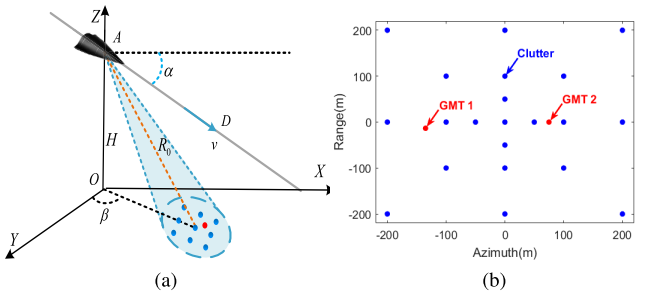


Fig. 10. Schematic of geometry correction.

TABLE III
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
Carrier-frequency	10 GHz	Equivalent squint angle	60 deg
Doppler bandwidth	4761 Hz	Subarray number	5
Height	20 km	PRF	1600 Hz
Pulse duration	10 us	Vertical velocity	1428 m/s
Diving angle	36.87 deg	Horizontal velocity	1904 m/s



(a) (b)

Fig. 11. Experiment sketch of point targets. (a) Geometry model. (b) Distribution of scattering points.

IV. SIMULATIONS AND EXAMPLES

To show the feasibility and superiority of the proposed scheme, we provide here three cases from different aspects.

A. Case 1: Point Targets

In this section, the simulations of point targets are performed to validate the effectiveness of the proposed scene imaging and GMTI scheme for HSV-DTHS-MC-SAR. The main system parameters are shown in Table III. Fig. 11 shows the scattering points sketch comprising SNR 0 dB, the CTV of two GMTs is 16 m/s, and the ATV of the first and the second GMTs are 0 and 16 m/s, respectively.

The echo is first processed by RC processing, and pulse-compressed results are shown in Fig. 12(a). After the LRWC, the profiles of all scattering points have been basically aligned, and the results of cross-range-dependent RCMC as shown in Fig. 12(b).

After the coarse-focusing preprocessing, the imageries of scattering points are shown in Fig. 13. With the azimuth FT processing, the profiles of the second GMT overlap with the surrounding stationary clutter, the image as shown in Fig. 13(a). With azimuth square CFT or cubic CFT, all scattering points are coarsely focused, two GMTs are compressed

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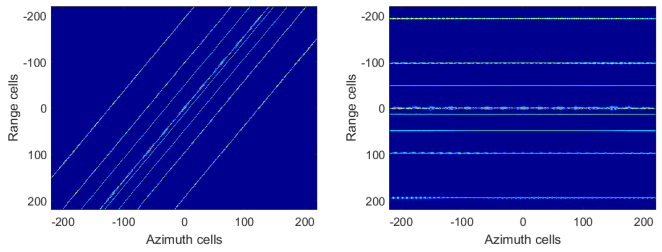


Fig. 12. Cross-range-dependent RCMC. (a) Without the LRWC. (b) After the LRWC.

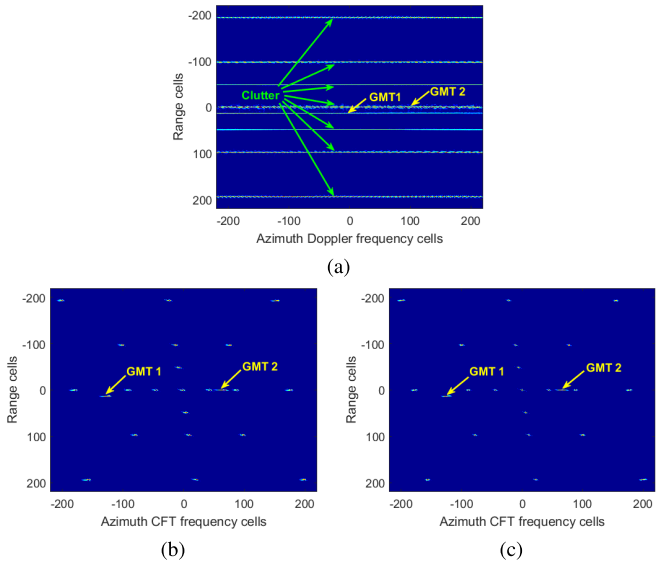


Fig. 13. Coarsely focused results. (a) After FT. (b) After square CFT of CFT-accurate method. (c) After the cubic CFT of the proposed method.

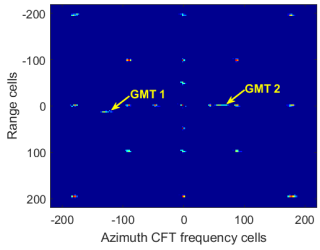


Fig. 14. Corrected image.

in several CFT frequency units and are therefore separable, and the images as shown in Fig. 13(b) and (c). Significantly, the square CFT cannot correct the profile distortion caused by the third-order phase, especially in the case of HSV-DTHS-MC-SAR. Fortunately, the cubic CFT not only implements the azimuth compression but also corrects the profile distortion caused by the third-order phase. Moreover, the geometry with inverse projections is used to inhibit distortion, and a "shift image" in Fig. 13(c) is changed into a "square image" as Fig. 14.

Fig. 15 summarizes the beamforming of different clutter suppression methods, and the real GMT direction is represented by the red arrow. It illustrates that the proposed method not only enjoys the highest amplitude in the GMT direction

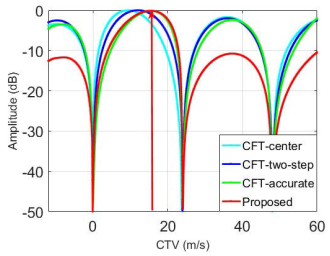


Fig. 15. Beamforming.

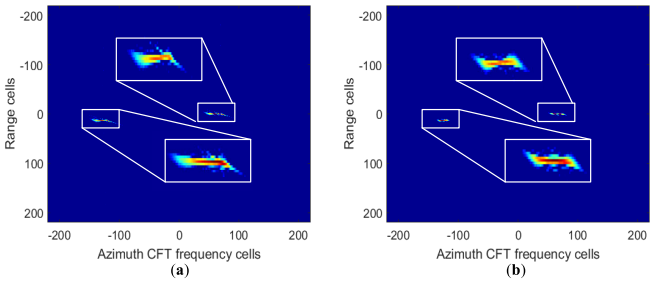


Fig. 16. Images after clutter rejection. (a) Square CFT of CFT-accurate method. (b) Cubic CFT of the proposed method.

but also has the lowest sidelobe amplitude. The reason is that the GMT steering vector mismatch arising from the imprecise CTV exists in the CFT-center and CFT-two-step methods. While the CFT-accurate method avoids GMT steering vector mismatch, its LCMV beamformer with the higher sidelobe may affect clutter suppression performance.

With the clutter cancellation, two GMTs and their magnified imageries are illustrated in Fig. 16. In Fig. 16(a), the image distortion is present, because the cubic phase error is still not removed during square CFT processing [6], [18]. The proposed method recovers the satisfactory GMT profiles, as it relieves cubic phase error and enjoys matched beamformer, the result is shown in Fig. 16(b) [7], [12].

Fig. 17 summarizes the accurate focusing processer of the first GMT. To be specific, the time-domain profile after coarsely focused is not aligned, and the image is illustrated in Fig. 17(a), and results after SOKT, accurate RWC, and cubic phase compensation are shown in Fig. 17(b)–(d). The sidelobe of the first GMT is low and geometry distortion is corrected in Fig. 17(d). Then, we further analyze the focusing result in Fig. 17(d), the slice plots with 2-D and azimuth cell are shown in Fig. 18(a) and (b), respectively. One notices that the coupling between the main lobe and sidelobe is removed in the first figure, and the peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) in the second figure are 0.17 and 0.14 dB higher than the theoretical values. Therefore, the focusing result is quantified and basically satisfactory.

Finally, we have quantified the performance gain and compared it to the existing methods, such as ISTAP, CFT-center, CFT-two-step, and CFT-search methods. With the simulation parameters in Table III, the comparison results between the proposed method and the existing methods are summarized in Tables IV and V. We shall further discuss the performance gain based on the SCNR, parameter estimation, and algorithm complexity, respectively.

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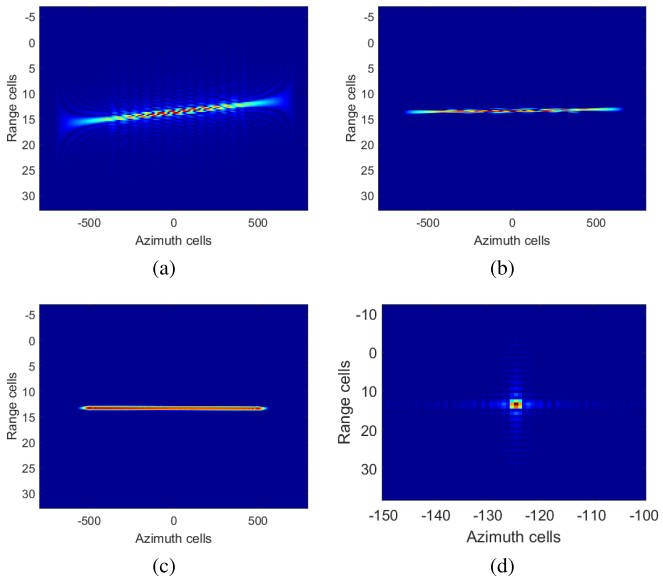


Fig. 17. Finer-focusing processing of the first GMT. (a) After coarsely focusing on the time domain. (b) After SOKT. (c) After accurate RWC. (d) After cubic phase compensation.

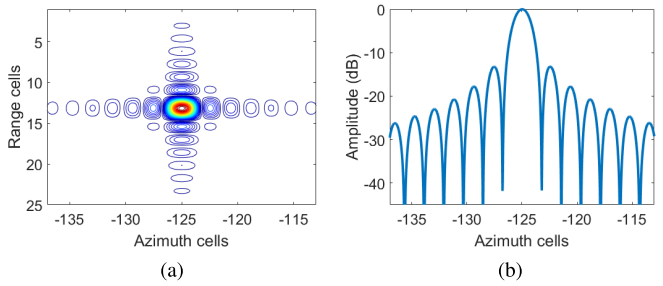


Fig. 18. Responses of accurate focused image. (a) Two-dimensional. (b) Azimuth.

TABLE IV
COMPARISON OF DIFFERENT METHODS

Method	SCNR (dB)	CTV error (m/s)	ATV error (m/s)
ISTAP [51]	21.24	3.27	3.64
CFT-center [18]	27.21	2.58	/
CFT-two-step [6]	30.55	1.96	0.23
CFT-search [13]	31.06	0.18	0.17
Proposed	36.29	0.11	0.12

TABLE V
ALGORITHMIC COMPLEXITY DURING CLUTTER SUPPRESSION

Method	Search	Algorithm complexity
ISTAP [51]	2-D search	$O(N_{nr}N_{na}N_{one})$
CFT-center [18]	/	$O(N_{one})$
CFT-two-step [6]	Coarse-search	$O(N_{nr}^{coarse}N_{one})$
CFT-search [12]	Finer-search	$O(N_{nr}N_{one})$
CFT-modified-search [13]	Two-step search	$O(N_{nr}/(N_{nr}^{coarse}+1)N_{one})$
Proposed	/	$O(N_{one})$

1) SCNR: The proposed scheme enjoys the best SCNR compared to the existing methods. Generally speaking, the proposed scheme with matched LCMV-QPC improves the beamformer matching and reduces the beamformer sidelobe, thus enhancing the GMT accumulation and reducing the clutter amplitude. The ISTAP method suffers from energy reduction, especially in the case of the limited DOF of the HSV platform. The CFT-center and CFT-two-step methods possess beamformer mismatch and thus result in GMT energy loss. While the CFT-search method frees the steering vector mismatch, has higher clutter amplitude caused by using the higher sidelobe of LCMV beamforming, and thus obtains the lower SCNR compared to the proposed method.

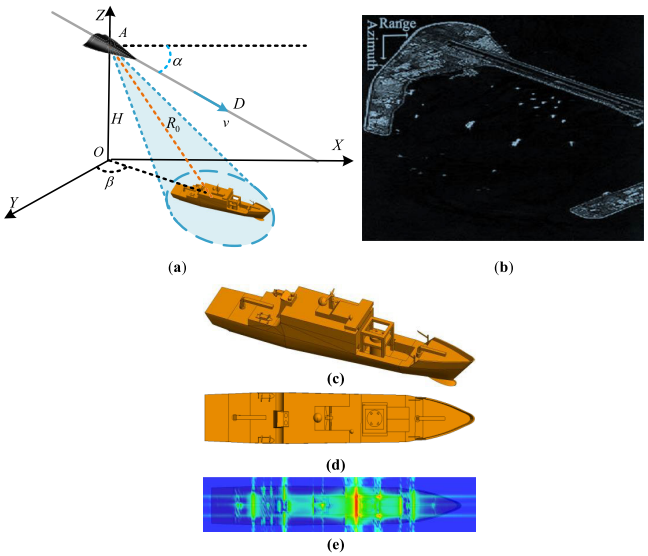


Fig. 19. Experiment sketch of surface target and extended harbor scene. (a) Geometry model. (b) Reference harbor scene. Surface target configuration with (c) 3-D model of the ship, (d) 2-D model of the ship, and (e) RCS of the ship.

2) Parameter Estimation: With the highest SCNR and the lower false alarm probability (FAP), the velocity estimation of the proposed scheme outperforms the existing three methods. To be specific, the parameter estimation performance of the ISTAP and CFT-center methods degrades significantly due to the lower SCNR. The estimation error of the CFT-two-step method also increases to varying degrees, due to the higher FAP arising from the finer search during GMT imaging and unwanted integration reduction of GMT local profiles caused by steering vector mismatch [7]. The performance of the CFT-search method is slightly inferior because its finer-search increases the FAP during clutter suppression and GMT imaging.

3) Algorithm Complexity: Table V lists the algorithmic complexity of the clutter suppression stage of existing methods, since most of the computation time is spent in this stage [13]. Also, N_{na} and N_{nr} denote the search numbers of along-track and cross-track parameters, N_{sa} and N_{sr} are the sampling numbers along the azimuth-time and range-time, $N_{\mathrm{nr}}^{\mathrm{coarse}}$ represents the coarse-search number of cross-track parameter. N_{one} is the algorithmic complexity of clutter suppression without parameter search, and $N_{\mathrm{one}} = N_{\mathrm{sr}}N_{\mathrm{sa}}^3N^3$. One can see that the proposed scheme has the lowest algorithmic complexity because it avoids the parameter search in existing methods. To be specific, the algorithmic complexity of the

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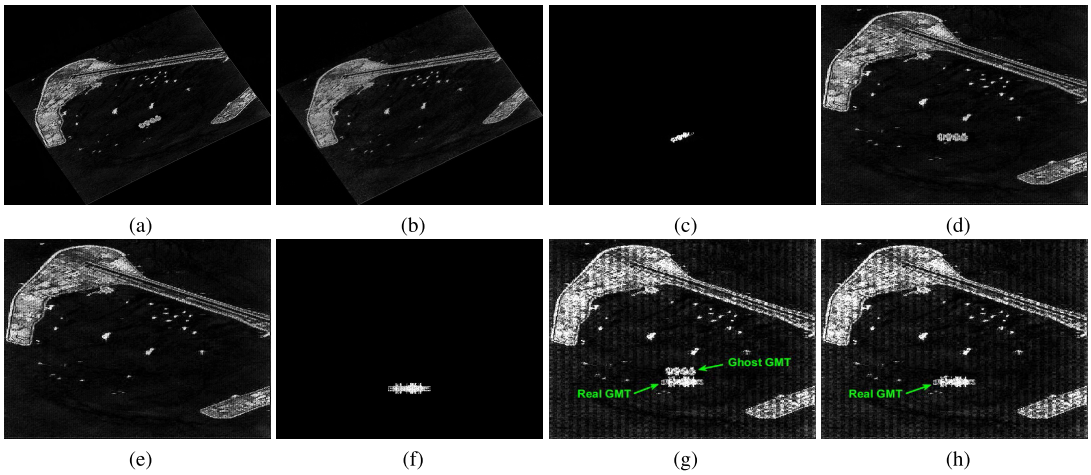


Fig. 20. Results of surface target and extended scene. (a) SAR scene precessing without ghost GMT suppression. (b) SAR scene precessing with ghost GMT suppression. (c) Clutter suppression and GMT extraction. (d) Geometry correction of SAR scene without ghost GMT suppression [13]. (e) Geometry correction of SAR scene with ghost GMT suppression. (f) Geometry correction and GMT finer-focusing. (g) GMT localization in SAR scene with ghost GMT [13]. (h) GMT localization in SAR scene without ghost GMT.

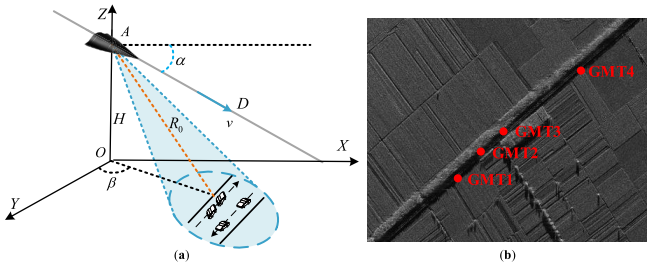


Fig. 21. Experiment sketch of multiple GMTs and ground SAR scene. (a) Geometry model. (b) Real GMT location in reference ground scene.

TABLE VI
GMT VELOCITY

GMTs	CTV (m/s)	ATV (m/s)
GMT1	16	0
GMT2	16	16
GMT3	-16	-16
GMT4	-8	-8

ISTAP method increases significantly, due to that the 2-D parameter search requires significant computation time and resource. After that, the CFT-search method, with the one-dimension search, has lower algorithmic complexity compared to the ISTAP method. Combining specific search operators during clutter suppression, the algorithmic complexities of the CFT-modified-search and CFT-two-step methods are further reduced. While the algorithmic complexity of the CFT-center method is the same as the proposed method in the clutter suppression stage, its performance of parameter estimation and beamformer matching deteriorates to some extent.

B. Case 2: Surface Target and Extended Harbor Scene

Taking a ship comprising 16 m/s CTV as a surface moving target, the experiment of target and extended harbor scene process is performed. The experiment parameters are as listed in Table III. The experiment sketch is shown in Fig. 19, the geometry model is sketched in Fig. 19(a), the reference harbor scene is shown in Fig. 19(b), and the configurations of 3-D and 2-D ship are illustrated in Fig. 19(c) and (d). With the FEKO 2018, the RCS of the ship is illustrated in Fig. 19(e), corresponds to that sketched in Fig. 19(d).

The results of surface targets and extended scene imaging are summarized in Fig. 20. When a GMT is not suppressed in SAR scene processing, there is a ghost GMT in the recovered scene image, and the result is shown in Fig. 20(a). To inhibit the side-effect of ghost GMT, the scene recovery and GMT suppression with "customized" beamforming is performed, which is capable of recovering the scene and placing the null in the expected GMT direction, as shown in Fig. 20(b). Besides, the result of clutter suppression and GMT extraction is shown in Fig. 20(c), one notices that a coarse-focusing GMT is separated from the SAR scene. Using geometry and inverse projections to correct the distortion in Fig. 20(a) and (b), the "square image" are obtained as shown in Fig. 20(d) and (e), very close to the reference harbor scene in Fig. 19(b), which indicates that the geometry distortion is basically corrected. Also, the finer-focusing image of a ship is retrieved in Fig. 20(f), due to that the high-order phase with GMT's 2-D motion parameters are estimated and compensated. To further show the side-effect of ghost GMT, we provide the comparison of GMT localization in SAR scene results in Fig. 20(g) with ghost GMT and that shown in Fig. 20(h) without ghost GMT.

C. Case 3: Multiple GMTs and Ground SAR Scene

Due to that the HSV-SAR/GMTI data are rare and not publicly available, we apply the simulation of scene reflectivity and GMTs to try to validate the superiority of the proposed scheme, the main system parameters and the velocity of GMTs are listed in Tables III and VI, and the experiment sketch is shown in Fig. 21.

The results of multiple GMTs and ground SAR scene imaging are summarized in Fig. 22. When a GMT is not

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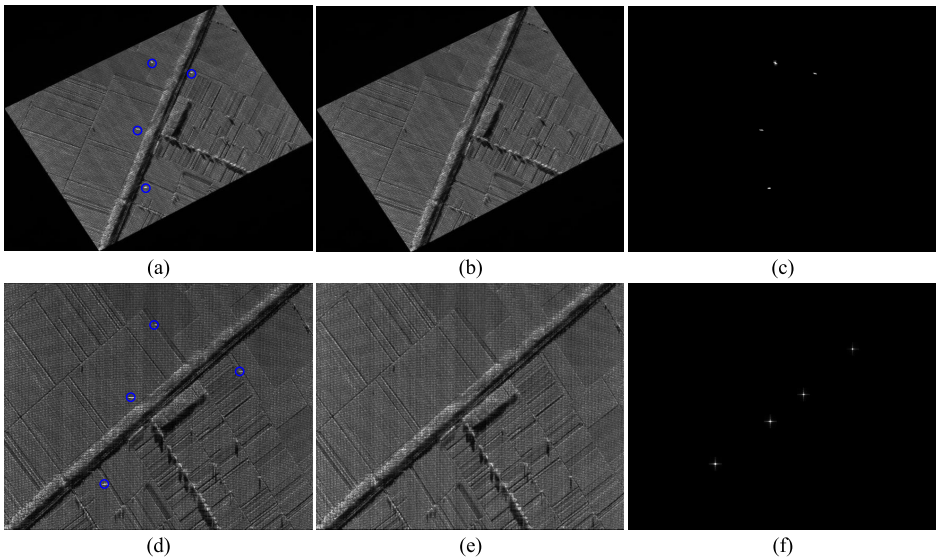


Fig. 22. Results of multiple GMTs and ground SAR scene. (a) SAR scene precessing without ghost GMT suppression. (b) SAR scene precessing with ghost GMT suppression. (c) Clutter suppression and GMT extraction. (d) Geometry correction of SAR scene with ghost GMT. (e) Geometry correction of SAR scene without ghost GMT. (f) Geometry correction and GMT finer-focusing.

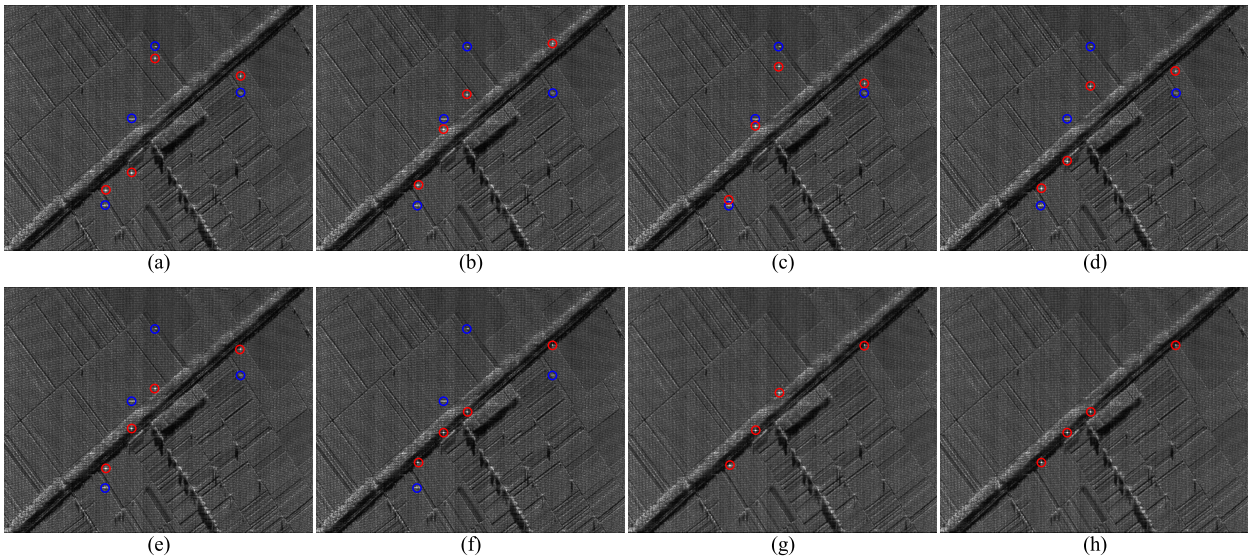


Fig. 23. Results of multiple GMTs and real SAR scene. (a) AMF [32]. (b) WAMF [33]. (c) CFT-center [18]. (d) CFT-two-step [6]. (e) CFT-modified-search [13]. (f) CFT-accurate [7]. (g) LCMV with ghost GMT suppression. (h) Proposed methods.

suppressed in SAR scene processing, there is a ghost GMT in the recovered scene image, and the result is shown in Fig. 22(a). To inhibit the side-effect of ghost GMT, the scene recovery and GMT suppression with "customized" beamforming is performed, which is capable of recovering the scene and placing the null in the expected GMT direction, as shown in Fig. 22(b). Besides, the result of clutter suppression and GMT extraction is shown in Fig. 22(c), one notices that a coarse-focusing GMT is separated from the SAR scene. Using geometry and inverse projections to correct the distortion in Fig. 22(a) and (b), the "square image" are obtained as shown in Fig. 22(d) and (e), very close to the reference ground scene in Fig. 21, which indicates that the geometry distortion is basically corrected. Also, the finer-focusing image of a ship is retrieved in Fig. 22(f), due to that the high-order phase with GMT's 2-D motion parameters are estimated and compensated.

Moreover, in order to illustrate the performance of GMT localization, the localization results of the difference methods are summarized in Fig. 23, where the ghost and real GMTs are marked by the blue and red circles. From Fig. 23(a), most real GMT positions of the AMF method are not marked on the road, because the estimation accuracy suffers from performance reduction when the high-order and channel phase errors are involved. While the WAMF method revises the search function and improves the CTV estimation and GMT location, the performance loss caused by phase error still exists to varying degrees, the result as shown in Fig. 23(b) [33], [34]. Then, the results of the CFT-center and CFT-two-step methods are shown in Fig. 23(c) and (d), which suffer from higher

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CTV estimation error and lower GMT location accuracy, and have unwanted GMT energy loss induced by steering vector mismatch. To cope with this problem, the CFT-search method with finer CTV search is performed, the results are shown in Fig. 23(e), and most real GMT positions are marked on or beside the road. However, we noticed that one GMT location is away from the road. The reason is that the lower SCNR and the higher FAP degrade the accuracy of ATV and CTV. To reduce the FAP and improve the location accuracy, the CFT-accurate method obtains satisfactory results as shown in Fig. 23(f). Significantly, the above methods do not consider the influence of ghost GMTs, so there is a possibility of confusion between real and ghost localization results. Fig. 23(g), the LCMV with ghost GMT suppression, which generates nulls in the GMT direction during the scene processing, inhibits the confusion of real and ghost GMTs. However, the LCMV beamforming, with its higher and irregular side-lobes, has lower SCNR, and thus leading to performance reduction of parameter estimation and location for some of the GMTs. Finally, in combination with lower beamformer sidelobe and satisfactory ghost GMT cancellation, the localization result of the proposed scheme is shown in Fig. 23(h), extremely close to that illustrated in Fig. 21. Therefore, the clutter suppression and GMT localization of the proposed scheme are reasonably feasible and superior. While the localization accuracy of the Dreamp-SATP method is close to the proposed scheme, it operates in a side-looking mode and is difficult to directly deal with HSV-DTHS-MC-SAR.

V. Conclusion

Incorporated with an IERM, this article explores a simultaneous scene imaging and GMTI scheme with multiple beamforming for HSV-DTHS-MC-SAR. The RC, LRWC, and azimuth cubic CFT are first performed to recover the coarse-focusing imageries of scene plus GMTs, thus mitigating the side-effect of GMT ambiguity. Then, in combination with dynamic-EASDB, the CTV estimation method is adopted that permits the vector steering mismatch to be inhibited. After that, the ghost GMTs in recovered SAR imageries can be effectively inhibited by "customized" beamforming during SAR scene processing. Moreover, we derive the GMT processing of HSV-DTHS-MC-SAR, including clutter suppression and GMT imaging plus localization. Also, a clutter suppression with a matched LCMV-QPC method improves the matching between the main-beam peak and GMT, enhances GMT accumulation, and improves the SCNR. Finally, the geometry and inverse projections are adopted to correct the distortion.

In the future, we shall further modify our work to tackle the HSV's complex orbit with acceleration, and the forward-looking SAR should be studied to deal with the left-right ambiguity problem with limited channel numbers. Moreover, the channel imbalance is also one of the hot topics for future research.

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